INCH-POUND

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MILITARY SPECIFICATION

# CUTTER, DELAY, PROPELLANT ACTUATED, M21 (2 SEC) AND M22 (10 SEC)

*This specification is approved for use by the Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.*

## 1. SCOPE

**1.1 Scope.** This specification covers the requirements for the M21 (2 SEC) and M22 (10 SEC) Propellant Actuated Cutter (see 6.1)

## 2. APPLICABLE DOCUMENTS

### 2.1 Government documents.

**2.1.1 Specifications and standards.** The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplements thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

| | |
|---|---|
| MIL-A-2550 | Ammunition, General Specification For |
| MIL-W-5625 | Webbing, Textile, Nylon, Tubular |
| MIL-S-13572 | Spring, Helical, Compression and Extension |
| MIL-C-13739 | Composition, Delay |
| MIL-P-20444 | Primer, Percussion, M42 Parts for Loading, Assembling and Packing |

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Standardization Branch (Code 8420), 101 Strauss Avenue, Indian Head, MD 20640-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by sending a letter.

AMSC N/A FSC 1377

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| | |
|---|---|
| MIL-P-22264 | Powder, Ignition, Gasless, A-1A |
| MIL-P-25732 | Packing, Preformed, Petroleum Hydraulic Fluid Resistant, Limited Service at 275°F |
| MIL-W-83420 | Wire, Rope, Flexible, for Aircraft Control |

STANDARDS

MILITARY

| | |
|---|---|
| MIL-STD-105 | Sampling Procedures And Tables For Inspection By Attributes |
| MIL-STD-109 | Quality Assurance Terms And Definitions |
| MIL-STD-129 | Marking for Shipment and Storage |
| MIL-STD-331 | Fuse and Fuse Components, Environmental and Performance Tests for |
| MIL-STD-413 | Visual Inspection Guide for Elastomeric O-rings |
| MIL-STD-453 | Inspection, Radiographic |
| MIL-STD-810 | Environmental Test Methods and Engineering Guidelines |
| MIL-STD-1168 | Lot Numbering of Ammunition |
| DOD-STD-2101 | Classification of Characteristics |

(Unless otherwise indicated, copies of federal and military specifications and standards are available from Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

**2.1.2 Other Government documents, drawings, and publications.** The following Government drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation (see 6.2).

DRAWINGS

Naval Surface Warfare Center, Indian Head (CAGE Codes 19200 and 19203)

| | |
|---|---|
| DL8875978 | Cutter, Propellant Actuated, 2 Second Delay, M21 |
| DL8875979 | Cutter, Propellant Actuated, 10 Second Delay, M22 |
| 6261283 | Box, Fiberboard (Inner), For Cutter, Delay, Propellant Actuated, MLU-58/B, M21, and M22 |



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| | |
|---|---|
| 6261284 | Box, Fiberboard, Packing (Outer), Ammunition for MLU-58/B, M21, and M22 Cutter |
| 6261285 | Box, Wood, Packing, Ammunition (Outer), For Cutter, Delay, Propellant Actuated, MLU-58/B, M21, and M22 |
| 8873322 | Cutter Tester |

(Application for copies should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Attn: Data Control Branch (Code 0342), Indian Head, MD 20640-5035.)

**2.2 Non-Government publications.** The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

| | |
|---|---|
| A 108 | Steel Bars, Carbon, Cold Finished Standard Quality |
| A 276 | Stainless and Heat-Resisting Steel Bars and Shapes |
| B 211 | Aluminum Alloy Bars, Rods and Wire |
| B 221 | Aluminum Alloy Extruded Bars, Rods, Wire, Shapes, and Tube |

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

**2.3 Order of precedence.** In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

## 3. REQUIREMENTS

**3.1 First article.** When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.4) in accordance with 4.6.

**3.2 General.** The cutters shall comply with all requirements on drawings listed on DL8875978 or DL8875979.



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**3.3 Materials.** All materials used in the manufacture of the cutters shall conform to the specifications referred to on the respective drawings unless specific approval in writing covering a departure has been obtained from the cognizant Navy design activity prior to manufacture. When alternate methods of manufacture are specified in the drawings, the bidder's selections shall be clearly stated in the proposal. Physical properties required shall be uniform throughout the part within specified tolerances. Requirements for all alloy steel parts shall be achieved by quenching and tempering (see 4.3).

**3.4 Primary components.** For the purpose of this specification, the ignition material (MIL-P-22264), the primer (MIL-P-20444), the delay mix (MIL-C-13739), and the output charge (Drawing 838AS115 are defined as primary components. Only primary components from a single lot shall be used in a production lot of cutters. One primary component production lot may be used in more than one cutter production lot.

**3.5 Primer.**

**3.5.1 Primer x-ray.** All primers shall be x-rayed and screened prior to assembly into the primer retainer (see 4.8.4).

**3.5.2 Primer sensitivity.** The primer shall fire when drop tested at an energy level of 26-inch-ounces in a primer retainer assembly (see 4.8.5).

**3.6 Spring load and spring embrittlement.** The springs shall not break and shall meet the requirements of Drawing 8875988 when tested as specified in 4.8.6 and 4.8.7.

**3.7 Corrosion resistance.** No component (except for the delay body, Drawing 6261279; washer (retainer), Drawing 6261280; and primer retainer, Drawing 8875985) shall exhibit white corrosion products after one hundred hours of salt spray testing. The delay body, washer, and primer retainer shall not exhibit white corrosion products after 24 hours of salt spray testing (see 4.8.8).

**3.8 Leakage.**

**3.8.1 Airtightness.** All knife delay assemblies shall be leak tested and shall not exhibit a leakage rate in excess of $1 \times 10^{-4}$ cc of air per second at a pressure differential of one atmosphere (see 4.8.9).

**3.8.2 Waterproofness.** The knife delay assembly shall not show any evidence of leakage when tested as specified in 4.8.10.

**3.9 Lanyard assembly.** The lanyard assembly shall withstand a tensile load of 200 pounds minimum without separation of parts (see 4.8.11).

**3.10 Radiographic examination.** After completion of assembly, all cutters shall be radiographically examined for proper assembly and presence of components. Any evidence of misassembly or missing parts shall be cause for rejection of that assembly (see 4.8.12).

**3.11 Staking.** The staking between the firing pin assembly and case shall withstand a breakaway torque of 25 inch-pounds without turning or loosening of the components (see 4.8.13).



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**3.12 Vibration.** The cutter shall withstand the vibration test without evidence of any disintegration of the primer, delay column, propellant, or damage of any component (see 4.8.14).

**3.13 Non-withdrawal and withdrawal of cotter pin.** The safety cotter pin shall not withdraw from the cutter when a pull force of 10 pounds minimum is applied to the pin. The pin shall be withdrawn when a pull force of up to 30 pounds maximum is applied to the pin (see 4.8.15).

**3.14 Functional.**

**3.14.1 Initiation.** The firing pin shall not be released when a force of 10 pounds is gradually applied to the lanyard assembly. The cutter shall function when the firing pin is released. The firing pin shall release when a force equivalent to dropping a 20-pound weight from a height of 3 ± 1/3 inches is applied to the lanyard assembly (see 4.8.16).

**3.14.2 Cutting.** The cutter shall completely sever two reefing lines when functioned (see 4.8.17).

**3.14.3 Delay time.** The delay time shall be as indicated in table I when tested in accordance with 4.8.17 (see 6.6.2).

**3.14.4 Functioning over temperature range.** The cutter shall function satisfactorily without failure (see 6.6.1) throughout the temperature range of −65 to 165°F when tested in accordance with 4.8.17.

TABLE I. Delay time.

| Temperature (°F) | Delay time (seconds) | |
|---|---|---|
| | M21 | M22 |
| −65 | 2.3 ± 0.7 | 11.0 ± 2.0 |
| 70 | 2.0 ± 0.4 | 10.0 ± 0.1 |
| 165 | 1.8 ± 0.6 | 9.5 ± 1.5 |

**3.14.5 Functioning at high setback.** The cutter shall function satisfactorily, as evidenced by complete severing of the reefing lines when a shock force of 700 g minimum is applied to the cutter (see 4.8.18).

**3.14.6 Misfire.** There shall be no misfires.

**3.15 Workmanship.** The workmanship requirements are as specified on the drawings, specifications, and the following. Workmanship and finish shall be in accordance with the highest grade practice used in manufacturing military weapons. Finished items and parts shall not exhibit poor material and processing such as seams, laps, laminations, cracks, visible steps, sharp edges, nicks, scratches, butts, deformations, and missing operations that may affect assembly, serviceability, functioning, operation, appearance, or safety. Fins and other extraneous metal shall be removed from cast or forged



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parts. Hammering to shape, salvage operations (including repair by welding), or other similar practices shall not be permitted without prior approval of the contracting officer. Surface coatings shall be continuous except for a few light scratches not exposing base material.

**3.15.1 Metal defects.** All components shall be free from cracks, splits, cold shuts, inclusions, incomplete or damaged threads, porosity, or any similar defect.

**3.15.2 Burr.** No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

**3.15.3 Foreign matter.** No part or assembly shall contain chips, dirt, grease, rust, corrosion, or other foreign matter.

## 4. QUALITY ASSURANCE PROVISIONS

**4.1 Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements (see 6.2).

**4.1.1 Responsibility for compliance.** All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

**4.1.2 Quality assurance terms and definitions.** Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

**4.1.3 Inspections.** Inspections shall be in accordance with MIL-A-2550.

**4.2 Classification of inspections.** Inspections shall be in accordance with MIL-STD-2550, except as specified herein. Inspections specified herein shall be classified as follows:

a. Material inspection (see 4.3)
b. First article inspection (see 4.6)
c. Quality conformance inspection (see 4.7)



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**4.3 Material inspection.** Material inspection shall consist of certification supported by the verifying data that the materials listed in table II, used in fabricating the cutters, are in accordance with the applicable specification or requirement prior to such fabrication (see 6.3).

TABLE II. Material inspection.

| Material | Requirement paragraph | Applicable specification(s) |
|---|---|---|
| Aluminum | 3.3 | ASTM B 211, ASTM B 221 |
| Ignition composition | 3.3 | MIL-P-22264 |
| O-ring | 3.3 | MIL-P-25732 |
| Spring steel wire | 3.3 | MIL-S-13572 |
| Steel | 3.3 | ASTM A 108, ASTM A 276 |
| Delay composition | 3.3 | MIL-C-13739 |
| Primer | 3.3 | MIL-P-20444 |
| Steel cable | 3.3 | MIL-W-83420 |

**4.4 Inspection conditions.** Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in MIL-STD-810.

**4.5 Classification of characteristics.** The characteristics verified by the tests and examinations herein are classified as critical, major, or minor in accordance with DOD-STD-2101. Tests and examinations that verify critical characteristics are identified by the symbol (C) and major characteristics by the symbol (M). The number following the classification symbol indicates the serial number of the test or examination. Tests and examinations not annotated with a classification code are classified as minor.

**4.6 First article inspection.** First article inspection shall be performed after award of contract and prior to production. After inspection and provisional acceptance at the source, sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract. First article inspection shall be performed on sample units which have been manufactured in the same manner, using the same materials, equipment, processes, and procedures as to be used in regular production. All parts and materials including packaging and packing shall be the same as used for regular production and shall be obtained from the same source of supply (see 6.2 and 6.4).

**4.6.1 Sample size.** Unless otherwise specified, the first article shall consist of 28 assembled units, 10 complete sets of components and subassemblies (that is, 10 each of every component and 10 each of the lanyard and firing pin assemblies and 10 knife delay assemblies), and one set of metal components, without coating or protective finish, for dimensional and visual examination.



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**4.6.2 Inspection routine.** The first article sample shall be subjected to the inspections specified in table III.

TABLE III. First article inspection.

| Inspection | Requirement para. | Test method para. | Number of samples to be inspected |
|---|---|---|---|
| Visual and dimensional inspection of components | 3.1 | 4.8.1 | 10 components |
| Hardness | 3.3 | 4.8.2 | 5 components |
| Spring load | 3.6 | 4.8.6 | 5 |
| Embrittlement of spring | 3.6 | 4.8.7 | 5 |
| Corrosion resistance | 3.7 | 4.8.8 | 3 components |
| Airtightness | 3.8.1 | 4.8.9 | 10 |
| Waterproofness of knife delay assembly | 3.8.2 | 4.8.10 | 10 |
| Lanyard assembly pull test | 3.9 | 4.8.11 | 10 |
| Radiographic | 3.10 | 4.8.12 | 28 |
| Staking | 3.11 | 4.8.13 | 28 |
| Vibration | 3.12 | 4.8.14 | 12 |
| Non-withdrawal and withdrawal of cotter pin | 3.13 | 4.8.15 | 21 |
| Firing pin release and non-release | 3.14.1 | 4.8.16 | 28 |
| Functioning over temperature range | 3.14.4 | 4.8.17 | 28 |
| Functioning at high setback | 3.14.5 | 4.8.18 | 7 |

**4.6.3 First article sample failure.** The first article sample shall meet all the requirements in table III. The Government reserves the right to terminate its inspection upon any failure of assembly or component in the sample to comply with the stated requirements. In the event of failure of any sample, the Government reserves the right to require the contractor to take correction action and submit a new sample or portion thereof.

**4.7 Quality conformance inspection.**

**4.7.1 Inspection of product.** Inspection of product shall consist of Group A inspection and Group D inspection, both done prior to delivery.

**4.7.1.1 Submission of product.** A lot shall consist of an identifiable group of assembled cutters containing metal parts assemblies produced by a homogeneous process and loaded with one lot of knife delay assemblies. Unless otherwise specified by the contracting activity, formation and identification of lots shall be in accordance with MIL-STD-105.

**4.7.1.2 Lot size.** A lot shall consist of an identifiable group of assembled cutters of not more than 1200 units.



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**4.7.1.3 Group A inspection.** Group A inspection shall consist of the inspections specified in table IV in the order shown (see 6.6.4).

TABLE IV. Group A inspection.

| Inspection | Requirement para. | Test method para. | Quantity tested | Accept/Reject criteria |
|---|---|---|---|---|
| Visual and dimensional examination | 3.1 | 4.8.1 | see 4.7.1.3.1 | 4.7.1.3.2 |
| Protective finish | 3.3 | 4.8.3 | per spec | per spec |
| Primer x-ray | 3.5.1 | 4.8.4 | all | No failures |
| Primer sensitivity | 3.5.2 | 4.8.5 | see 4.7.1.3.4 | No failures |
| Spring load | 3.6 | 4.8.6 | 10 | No failures |
| Embrittlement of springs | 3.6 | 4.8.7 | 10 | No failures |
| Airtightness | 3.8.1 | 4.8.9 | 10 | 4.7.1.3.2 |
| Waterproofness of knife assembly | 3.8.2 | 4.8.10 | 65 | 4.7.1.3.2 |
| Lanyard assembly pull test | 3.9 | 4.8.11 | all | 4.7.1.3.2 |
| Radiographic inspection of assembled cutters | 3.10 | 4.8.12 | all | 4.8.12 |
| Staking test | 3.11 | 4.8.13 | all | No failures |
| Withdrawal and non-withdrawal of cotter pin | 3.13 | 4.8.15 | 25 | 4.7.1.3.2 |
| Release and non-release of firing pin | 3.14.1 | 4.8.16 | all | 4.7.1.3.2 |

**4.7.1.3.1 Examination.** One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis (see 6.6.3) in accordance with the classification of each applicable drawing of DL8875978, DL8875979, or as contained herein, using the sampling plan and associated criteria specified in the contract. In addition to the characteristics shown on Drawings 8875978 or 8875979, examination for characteristics listed in table V shall also be made. All nonconforming material shall be rejected.

**4.7.1.3.2 Component or assembly failures.** Any components or assembled cutters failing to meet requirements shall be cause for rejection of that item. In the waterproofness of knife delay assembly test, if two failures occur, the lot is rejected. If only one failure is found, a second sample of 130 assemblies shall be tested. When a total of two defectives is encountered in the combined first and second sample, testing will be stopped and the lot shall be rejected. In the non-withdrawal and withdrawal of safety cotter pin test, if more than one failure occurs, the lot shall be rejected. If only one defective is found, a second sample of 50 shall be tested. If more than one defective is found in the second sample, the lot shall be rejected.



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TABLE V. Classification of assembly defects.

| Classification | Defect | Method of Inspection |
|---|---|---|
| Critical | | |
| 1 | All parts are present and assembled correctly | Visual/x-ray |
| Major | | |
| 101 | Assembly damaged | Visual |
| 102 | Stake inadequate | Visual |
| 103 | Marking clear and legible | Visual |
| Minor | | |
| None defined | | |

**4.7.1.3.3 Rejected lots.** If an inspection lot is rejected, the disposition of the lot shall be as directed by the contracting activity. In the event the contractor is directed to rework the lot, a written procedure for the rework shall be submitted to the responsible Government engineering activity for approval before commencing the rework. Such lots shall be separate from new lots and shall be clearly identified as reworked lots in accordance with MIL-STD-1168.

**4.7.1.3.4 Primer sensitivity sample.** The sample for primer sensitivity shall be in accordance with table VI.

TABLE VI. Primer sensitivity sample.

| Lot size | Primer sensitivity sample size |
|---|---|
| 3 to 40 | 5 |
| 41 to 110 | 10 |
| 111 to 800 | 35 |
| over 800 | 50 |

**4.7.1.4 Group D inspections.** Group D inspections shall consist of the tests specified in table VII and shall be made on sample units which have undergone and passed Group A inspection. The sample shall consist of 30 cutter assemblies selected from each lot, unless otherwise specified by the contracting activity (see 6.6.5).



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TABLE VII. Group D inspection.

| Inspection | Requirement paragraph | Test method paragraph | Quantity tested | Accept/Reject criteria |
|---|---|---|---|---|
| Vibration | 3.12 | 4.8.14 | 12 | 4.7.1.4.1 |
| Release and non-release for firing pin | 3.14.1 | 4.8.16 | 30 | 4.7.1.4.1 |
| Functional test over temperature range | 3.14.4 | 4.8.17 | 21 | 4.7.1.4.1 |
| Functional test at high setback | 3.14.5 | 4.8.18 | 9 | 4.7.1.4.1 |

**4.7.1.4.1 Test failure.** The cutter shall be classified as defective if it releases the firing pin under minimum force, fails to fire under maximum force, fails to meet specified time requirement, or fails to completely sever both reefing lines. Failure of any unit to comply with requirements of 3.12 and 3.14 shall be cause for lot rejection.

**4.7.1.4.2 Retest.** There shall be no retests of a failed lot unless directed by the contracting activity. The retest shall be done under direction of the responsible Government engineering activity.

**4.8 Tests and examinations.**

**4.8.1 Visual and dimensional examinations.** Visual examination shall be made for such defects as defective metal, burr, sharp corners, foreign matter, defective protective finish, marking missing, incorrect, or illegible, thread damaged or not full, holes, slots, radii or chamfer missing. Four power (4X) magnification may be used in performing visual examination for such defects. An appropriate gage or measuring device shall be used to check all other characteristics shown on the drawing. Inspection for rubber O-rings shall be in accordance with MIL-STD-413 and MIL-A-2550.

**4.8.2 Hardness.** Prior to application of protective finish, a hardness tester shall be used to determine the hardness values specified on the applicable drawing.

**4.8.3 Protective finish.** The procedures for protective finish shall be in accordance with the component drawing.

**4.8.4 Primer x-ray (C1).** Prior to assembly into the primer retainer, the primer shall be x-rayed with the side of cup perpendicular to the rays. The x-ray negative shall be identified by item nomenclature and lot number. After screening and indicating defective primers, the x-ray photographs shall be forwarded to the responsible Government engineering activity (see 6.3).

**4.8.5 Primer sensitivity.** Tests shall be made on primers assembled into primer retainers. The primer shall be crimped and sealed in the retainer in accordance with the assembly drawing. Drop test at 13 inches (adjusted) using a stainless steel ball weighing 1.94 $\pm$ 0.2 ounces in a primer sensitivity tester with a primer retainer holder approved by the contracting activity. The drop height of 13 inches shall be adjusted taking into consideration the actual weight of the ball so that an energy level of 26 inch ounces shall be achieved. All height measurements shall be made between the head of the firing pin and



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the bottom of the steel ball. The primer sensitivity test equipment shall be checked to ensure a center impact of the steel ball on the head of the firing pin.

**4.8.6 Spring load.** The spring shall be assembled in the test equipment and force applied until the spring is compressed to the larger height specified in Drawing 8875988. The applied force shall be recorded. Force shall be reapplied until the unit is compressed to the smaller height specified in Drawing 8875988 and the applied force recorded. Any spring exhibiting a compression force at either height outside the load range permitted by Drawing 8875988 shall be classified defective.

**4.8.7 Spring embrittlement.** The spring shall be held at each end and stretched until the minimum point of permanent distortion is reached. If the spring breaks before reaching that point, it shall be classified defective. Springs so tested shall be discarded.

**4.8.8 Corrosion resistance.** Corrosion resistance testing shall be performed as specified in MIL-STD-810.

**4.8.9 Airtightness.** Each knife delay assembly shall be tested in a dry gas leak tester.

**4.8.10 Waterproofness of knife assembly.** The knife delay assembly shall be placed under water in a vacuum chamber. The water shall contain the wetting agent Maccon NR (sodium alkyl aryl sulfonate) or equal. The chamber shall be evacuated to a minimum pressure differential of 5 psi. Any assembly emitting bubbles, at any frequency, from one or more points, shall be classified as defective and discarded. After subjection to test, the assembly shall be rinsed thoroughly in fresh running water to remove the wetting agent and then dried at room temperature.

**4.8.11 Lanyard assembly pull test.** The lanyard assembly shall be placed in the applicable test equipment wherein it shall be held at the firing pin release and through the lanyard eye. The specified load shall be applied. No dwell time is required. Any unit which separates shall be classified as defective.

**4.8.12 Radiographic examination.** Each assembled cutter shall be radiographically examined in accordance with MIL-STD-453 with the exception that a fully assembled cutter shall be substituted for the penetrameter. The cutters shall be positioned for the most revealing exposure with the long axis of the cutter perpendicular to the rays of the x-ray machine. All assemblies shall be examined by radiographic plates and any observable imperfections in components or assembly shall be cause for cutter rejection. All assemblies shall be identified with serial numbers prior to examination. The cutter serial numbers shall be in consecutive numerical order beginning with the number 001 in each production lot. The assemblies for radiographic examination shall be arranged or trays or boards in numerical order and each radiograph shall carry a permanent identification of the assemblies displayed thereon. Any missing serial numbers are to be identified on the x-ray. The radiographic plate identification shall include the complete lot number, as stamped on the cutters, and the span of serial numbers displayed. Radiographic plates of the entire production lot shall be forwarded to the activity conducting the ballistic acceptance testing (see 6.3). Film interpretation of the radiographic plates shall verify that the final assembly is properly assembled and includes all components. Defective assemblies found during x-ray review are to be marked on the x-ray prior to shipment and those defective items shall be removed from the production lot.



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**4.8.13 Staking (M102).** The assembled cutter shall be placed in a holding device in which the prescribed amount of torque can be applied to the retainer to verify that the staking between case and retainer meets the requirements of 3.11.

**4.8.14 Vibration.** The selected sample shall undergo transportation vibration testing as specified by procedure II in Test 401 of MIL-STD-331.

**4.8.15 Withdrawal and non-withdrawal of cotter pin.** The assembled cutter shall be firmly mounted in a holding fixture. A force equal to the minimum specified withdrawal force shall be applied, without impact, to the cotter pin so as to tend to remove the pin from the assembly. The force shall act in a direction parallel to the length of the pin and shall be applied at the eye. If the safety cotter pin is not withdrawn when the minimum force is applied, the maximum force shall be applied in the same manner as specified above. Any cotter pin which is withdrawn when the minimum force is applied or is not withdrawn when the maximum force is applied shall cause the assembly to be classified as defective.

**4.8.16 Firing pin release and non-release.** The firing pin assembly (Drawing 6261286) shall be secured in a fixture. A pull force shall be applied to the lanyard in the longitudinal direction in conjunction with the fixture which registers the amount of pull. If the force to release the firing pin is less than the specified minimum, the assembly shall be classified as defective.

**4.8.16.1 Firing pin release in functional test.** In conjunction with functioning over temperature range tests, the force required to release the firing pin shall be recorded on the firing record.

**4.8.17 Functioning over temperature range.** The cutter shall be mounted in the test fixture, Drawing 8873322. Two reefing lines (1000 pound tensile strength, 5/8 inch, MIL-W-5625) shall be passed through the hole in the cutter. One end of the reefing lines shall be anchored and a 50-pound load attached to the free ends of the reefing lines. The lanyard shall then be pulled to release the firing pin. A ballistic trace which records the force applied to the lanyard to release the firing pin and the delay time (see 6.6.2) shall be made for each firing.

**4.8.17.1 Temperatures.** The sample for functioning over the temperature range shall be divided into three equal groups and temperature conditioned as shown in table VIII. The units shall be tested within 10 minutes after removal from the conditioning chamber. When this time is exceeded, the sample unit shall be reconditioned for a minimum of 1 hour.

**4.8.17.2 Temperature conditioning.** Each cutter shall be conditioned for not less than 4 hours nor more than 24 hours. When reconditioning cutters at $-65°F$, all condensation shall be removed from the exterior surfaces before returning the cutter to the conditioning chamber.

**4.8.18 Functioning at high setback.** The cutter assembled to the test fixture (Drawing 8873322) shall be initiated by release of the firing pin. Midway through the delay firing time the specified shock shall be applied in the direction of the burning of the delay. Observation shall be made for complete functioning as evidenced by complete severing of the reefing lines. This test is performed at ambient temperature as indicated in table VIII.



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TABLE VIII. Conditioning requirements for functioning test.

| Temperature (°F) | Vibrated | Non-vibrated | High setback |
|---|---|---|---|
| First article inspection sample | | | |
| −65 to −70 | 4 | 3 | — |
| 65 to 70 | 4 | 3 | 7 |
| 160 to 165 | 4 | 3 | — |
| Group D sample | | | |
| −65 to −70 | 4 | 3 | — |
| 65 to 70 | 4 | 3 | 9 |
| 160 to 165 | 4 | 3 | — |

**4.9 Inspection of packaging.** The sampling and inspection of the packaging, packing, and container marking shall be in accordance with the requirements of section 5.

## 5. PACKAGING

**5.1 Packing and marking.** Packing and marking shall be Level A or B, as specified by the contracting activity (see 6.2). Level A shall be in accordance with Drawings 6261283 and 6261285. Level B shall be in accordance with Drawings 6261283 and 6261284.

**5.2 Marking (M103).** Marking shall be in accordance with MIL-STD-129 and the requirements of Drawing 6261284 and 6261285.

## 6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

**6.1 Intended use.** The cutter is used to cut nylon reefing line in cargo air drops.

**6.2 Acquisition requirements.** Acquisition documents must specify the following:

a. Title, number, and date of this specification

b. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.1.2)

c. Whether first article sample is required and, if so, the test activity (see 3.1 and 4.6)

d. Production lot size and test activity (see 4.1 and 4.7)



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e. Ballistic test sample, if other than 30 units (see 4.7.1.4)

f. Direction for shipping radiographs of primers before assembly and of the entire production lot along with ballistic sample to the activity conducting production lot acceptance tests (see 4.8.4, 4.8.12, and 6.3)

g. Level of packing and marking (see 5.1)

h. Items of data required for materials inspection and production lot (see 6.3)

i. That the safety precaution requirements of the "Contractor's Safety Manual for Ammunition, Explosives, and Related Dangerous Material," DOD 4145.26M are applicable. NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of "Ammunition and Explosives Ashore," OP 5 are applicable.

**6.3 Data requirements.** The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

| Ref. Paragraph | DID Number | DID Title | Tailoring |
|---|---|---|---|
| 4.3 | DI-MISC-80678 | Certification Data Reports | None |
| 4.8.4, 4.8.12 | DI-MISC-80653 | Test Reports | None |
| 6.3.1 | DI-MISC-80043 | Ammunition Data Card | None |

(The above DID's were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.)

**6.4 First article.** When a first article inspection is required, the contracting officer should provide specific guidance to offerors whether the items should be a first article sample or a sample selected from the first production items (see 3.1), and the number of items to be tested as specified in 4.6.1. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examination, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.



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**6.5 Hazard notice.** The cutter described herein and certain of its components are flammable or explosive and present hazards in manufacture, handling, storage, and shipment. The contractor should recognize these properties and take appropriate measures to guard and protect against fire, explosion, adverse environment, corrosive atmosphere, rough handling, and electrically induced incidents.

**6.6 Definitions.** For the purpose of this specification, the following definitions apply:

**6.6.1 Mechanical failure.** A mechanical failure is defined as any deformation or breakage of a part, the occurrence of which is other than a design function.

**6.6.2 Delay time.** Delay time is defined as the time elapsed from firing pin release on the ballistic record until the severance of the reefing lines.

**6.6.3 Defect.** A defect is any nonconformance of a characteristic with specific requirements.

**6.6.4 Group A inspection.** Nondestruction inspection of all items acquired or all samples from an inspection lot to demonstrate product compliance with contractual requirements. Group A inspections examine characteristics most affected by variations in production processes or skills and functions vital to successful completion of design mission.

**6.6.5 Group D inspection.** Destructive tests that consume all of the design service life. Articles tested in Group D inspections are considered completely expended and will not be issued for service use. Tests are performed on few samples based on production quantities.

**6.7 Subject term (keyword) listing.**

Explosive cable cutter
Parachute reefing line cutter
Reefing line cutter

**6.8 Changes from previous issue.** Marginal notations are not used to identify changes with respect to the previous issue because of the extensiveness of the changes.

Preparing activity
Navy - OS
(Project 1377-NE36)



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

## INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

| I RECOMMEND A CHANGE: | 1. DOCUMENT NUMBER | 2. DOCUMENT DATE (YYMMDD) |
|---|---|---|
| | MIL-C-46992C | 930921 |

**3. DOCUMENT TITLE**

CUTTER, DELAY, PROPELLANT ACTUATED, M21 (2 SEC) AND M22 (10 SEC)

**4. NATURE OF CHANGE** *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

**5. REASON FOR RECOMMENDATION**

**6. SUBMITTER**

| a. NAME (Last, First, Middle Initial) | b. ORGANIZATION | |
|---|---|---|
| c. ADDRESS (Include Zip Code) | d. TELEPHONE (Include Area Code)<br>(1) Commercial<br>(2) AUTOVON (If applicable) | 7. DATE SUBMITTED (YYMMDD) |

**8. PREPARING ACTIVITY**

| a. NAME | b. TELEPHONE (Include Area Code) (1) Commercial | (2) AUTOVON |
|---|---|---|
| COMMANDER, INDIAN HEAD DIVISION<br>NAVAL SURFACE WARFARE CENTER (CODE 8420) | 301-743-4358/4510 | 354-4358/4510 |
| c. ADDRESS (Include Zip Code)<br>101 STRAUSS AVENUE<br>INDIAN HEAD, MD 20640-5035 | IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:<br>Defense Quality and Standardization Office<br>5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466<br>Telephone (703) 756-2340 AUTOVON 289-2340 | |



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